# Lecture XI - Arena Seat Planning under Distancing Rules

Applied Optimization with Julia

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# Introduction

## Covid-19 Pandemic

### Challenges for Live Events

- Overall number of participants at events **was restricted**
- Certain **spacing between participants** had to be ensured
- Larger events required **vaccination certificates** for all

**Question: What are the main issues for the organizers?**

### Main Difficulties

- Organization of larger events **is costly**
- Even without a pandemic a **financial risk**
- **Administrative Burden** for vaccination certificates
- Reduced capacity is a **loss of revenue**
- Implementing and enforcing **distancing rules**
- Managing **different priorities** of groups

## Idea: Optimizing Seating Plans

### Background

- **Applications:** sport arenas, concert halls, movie theaters, lecture halls, etc.
- People from the same group are **seated together**
- Venues have **specific seating**, season tickets, VIPs, etc.

> ! **Important**
> Optimizing seating plans can help to **maximize revenue** while ensuring distancing rules and other constraints are met.

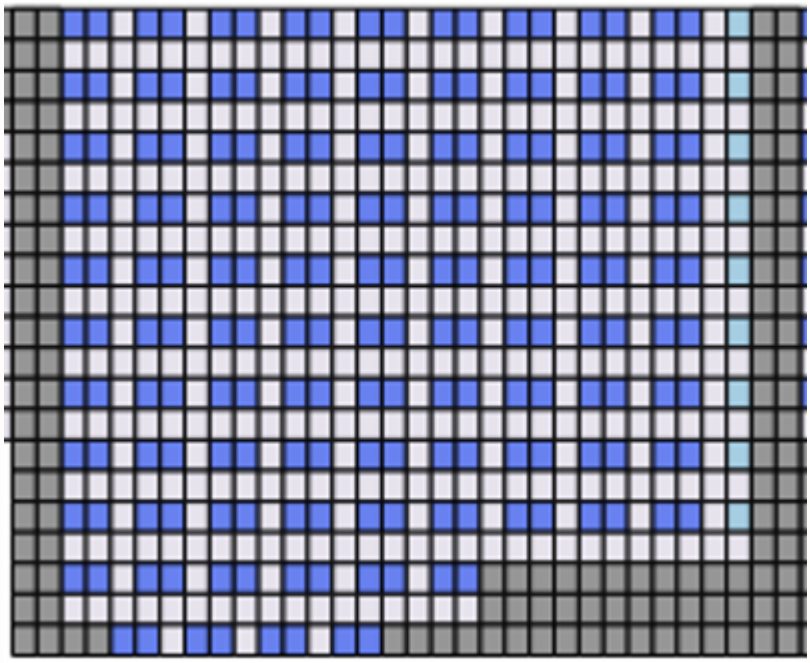
### Learning Goals

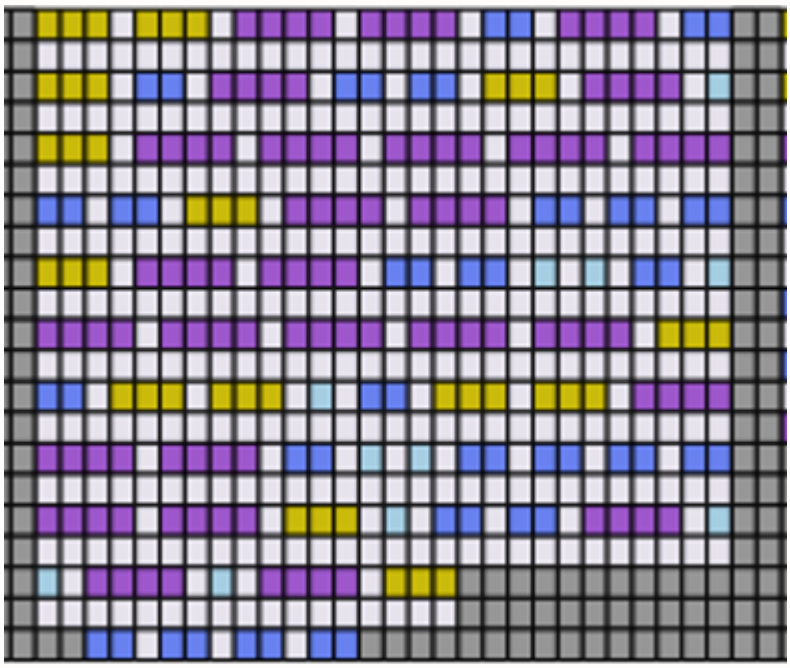
After this lecture, you will be able to:

- **Formulate** the arena seating problem as a binary program
- **Model** group placements with one binary variable per starting seat
- **Enforce** distancing rules with rectangle covering constraints
- **Distinguish** operational, tactical, and strategic seating decisions

# Problem Structure

## Example: Two different plans





## Different Approaches Possible

1. Operational
2. Tactical
3. Strategic

**Question: What are these approaches in general and how do they relate to arena seating?**

## Operational

- **Short-term**, day-to-day decisions
- Focused on **immediate execution**

**Question: What is an example of this approach?**

- Given **tomorrow's demand** of differently sized groups
- Score groups (importance, sponsors, VIP, season ticket,...)
- Assigning specific seats for **tomorrow's event**

## Tactical

- **Medium-term** planning (weeks to months)
- Bridges operational and strategic levels

**Question: What is an example of this approach?**

- Given distribution of **expected demand** for groups
- Score groups (importance, sponsors, VIP, season ticket,...)
- Plan seating arrangements for an **upcoming season**

### Strategic

- Long-term planning (months to years)
- Focus on overall goals and policies

**Question: What is an example of this approach?**

- Design one **fixed layout** that works well across different demand scenarios
- Maximize the overall **space utilization**
- Sell the resulting layout as a **standard product** on the market

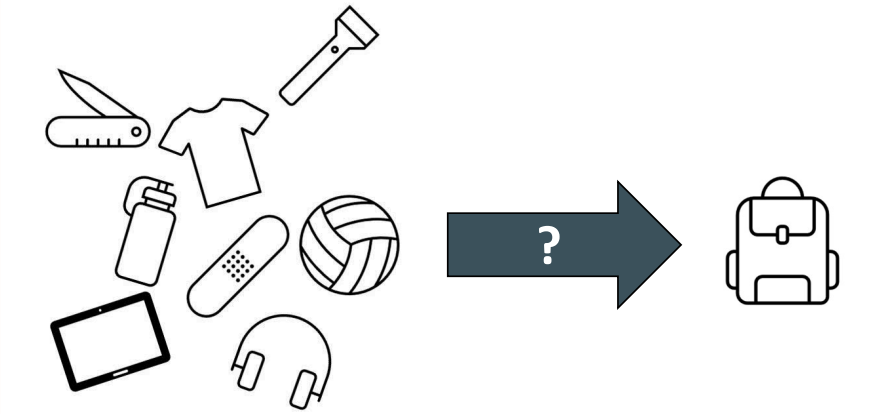
### Main Question

**Task:** Fill the seating area given distancing regulations and venue-specific constraints.

**Question:** Any ideas on how to approach this?

## Knapsack

### Knapsack Problem




- **Standard model in Operations Research**
- **Select items** from a pool under **capacity constraints**

## Knapsack Problem in 2D

| | c1 | c2 | c3 | c4 | c5 | c6 | c7 | c8 | c9 | c10 | c11 | c12 | c13 | c14 | c15 | c16 | c17 | c18 | c19 | c20 | c21 | c22 | c23 | c24 | c25 | c26 | c27 | c28 | c29 | c30 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| r1 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 12 | 12 |
| r2 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 12 | 12 |
| r3 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r6 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r7 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r8 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r9 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r10 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r11 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r12 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r13 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r14 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | 12 | 12 |
| r15 | 1 | 1 | 1 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 4 | | |
| r16 | 1 | 1 | 1 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| r17 | 1 | 1 | 1 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| r18 | 1 | 1 | 1 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| r19 | 1 | 1 | 1 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| r20 | 1 | 1 | 1 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |

- Now, items block space in 2D, as illustrated here

## Adaptation to Seating

- Horizontal dimension to place groups of participants
- Vertical dimension to ensure enough spacing between rows
- Maximize the “value” of the allocated groups
- Value can be the number of seats or a score

> **Note**
> Idea behind the model was developed by Dr. Matthes Koch.

## Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/arena.html

## Which Groups Did You Pick First?

**Question: How valuable is each group per seat?**

- Groups c and g yield **2 points per seat**
- Group e yields 1.25 points, all other groups exactly 1
- Grabbing the “densest” groups first is a **greedy strategy**

> **Note**
> Placing all groups would require 53 seats and yield 67 points — but the blocked seats and the distancing rules make this impossible.

### How Good Is Your Plan?

**Question: What is the best possible score?**

- The optimal seating plan reaches **52 points**
- Manual plans usually fall a few points short
- Greedy choices are a good start, but **combining** them optimally is hard

> **Tip**
> In the tutorial, we will model exactly this seating area and let the solver find the optimal plan **within seconds**.

## Model Formulation

### Sets?

**Question: What could be the sets?**

- $\mathcal{G}$ - Set of groups, indexed by $g$
- $\mathcal{R}$ - Set of rows, indexed by $r$
- $\mathcal{C}$ - Set of columns, indexed by $c$
- $\mathcal{C}_{g,r}$ - Available seats of row $r$ for group $g$, indexed by $c$

> **Note**
> $\mathcal{C}_{g,r}$ contains only **feasible starting seats**: a seat $c$ belongs to it only if the entire group $g$ fits into row $r$ from seat $c$ onward without crossing a blocked seat or the end of the row. This is why the set depends on both $g$ and $r$.

### Parameters?

**Question: What could be possible parameters?**

- $p_r$ - Maximal number of groups allowed in one row $r$
- $d_g$ - Required seats of a group $g$ in a row
- $h$ - Safety distance between groups sitting next to each other
- $b$ - Vertical safety distance between groups
- $v_g$ - Value of an allocation of the group $g$

## Variables & Objective

### Decision Variable?

> **Our goal is to:**
> Maximize the group values by filling the seating area given distancing regulations between groups and venue-specific constraints.

> **Tip**
> Each group is represented by **one binary variable**. We don't need to block each seat explicitly with a binary variable! The sets $\mathcal{C}_{g,r}$ make this possible, as variables only exist where the **whole group** actually fits.

## Decision Variable

> **We need the following sets:**
> - All the groups, $g \in \mathcal{G}$
> - All the rows, $r \in \mathcal{R}$
> - All the columns, $c \in \mathcal{C}$

**Question: What could be our decision variable?**

- $X_{g,r,c}$ - 1, if **first left seat** of $g$ is assigned to $r$ in $c$, else 0

## Objective Function?

> **Our main objective is to:**
> Maximize the group values by filling the seating area given distancing regulations between groups and venue-specific constraints.

**Question: How again are groups allocated?**

- By the allocation of the **first left seat** of a group to a row and column in the seating area

## Objective Function

> **We need the following parameters and variables:**
> - $v_g$ - Value of an allocation of the group $g$
> - $X_{g,r,c}$ - 1, if **first left seat** of $g$ is assigned to $r$ in $c$, else 0

**Question: What could be our objective function?**

$$\text{Maximize} \quad \sum_{g \in \mathcal{G}} \sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} v_g \times X_{g,r,c}$$

# Constraints

## Necessary Constraints

**Question: What constraints do we need?**

- Assign **each group only once**
- Restrict the **number of groups in each row**
- Ensure the **horizontal social distance**

- Keep the **vertical social distance**

## Assign Each Group Only Once?

! **The goal of this constraint is to:**
Ensure that each group is allocated only once in the entire seating area.

i **We need the following:**
- $X_{g,r,c}$ - 1, if first left seat of $g$ is assigned to $r$ in $c$, else 0
- $\mathcal{G}$ - Set of groups, indexed by $g$
- $\mathcal{R}$ - Set of rows, indexed by $r$
- $\mathcal{C}_{g,r}$ - Set of columns of row $r$ for group $g$, indexed by $c$

## Assign Each Group Only Once

**Question: What could be the constraint?**

$$\sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq 1 \quad \forall g \in \mathcal{G}$$

i **Note**
This "set packing" constraint ensures that a group is only **assigned once**.

## Restrict Groups Per Row?

! **The goal of this constraint is to:**
Ensure that the number of groups in each row does not exceed the maximum allowed number of groups.

i **We need the following:**
- $p_r$ - Maximal number of groups allowed in one row $r$
- $X_{g,r,c}$ - 1, if first left seat of $g$ is assigned to $r$ in $c$, else 0

## Restrict Groups Per Row

**Question: What could be the constraint?**

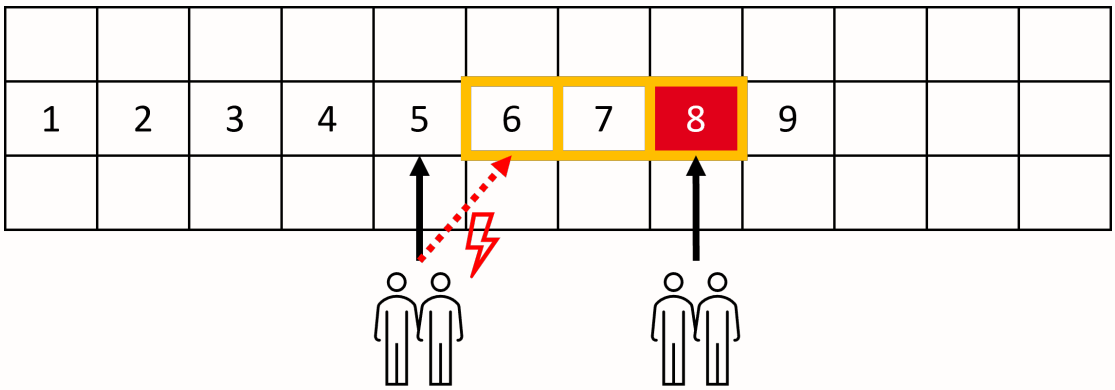
$$\sum_{g \in \mathcal{G}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq p_r \quad \forall r \in \mathcal{R}$$

i **Note**
We want to place as **many highly scoring groups as possible**, but people need to move to buy drinks or use the restroom. Depending on the venue, they should **not cross other groups** in the **same row**.

The last two
constraints are
somewhat tricky!

# Social Distancing

## Central Idea




> **Tip**
> Assume **one seat between groups must be kept empty**. If one group takes seat 8, it uses seats 8 and 9. We thus **cannot allocate** another group of size 2 to seats 6, 7 or 8.

## Horizontal Social Distance?

**Question: Any ideas how to implement this?**

> **The goal of this constraint is to:**
> Ensure that the horizontal social distance is maintained between groups.

> **We need the following:**
> - $X_{g,r,c}$ - 1, if first left seat of $g$ is assigned to $r$ in $c$, else 0
> - $d_g$ - Required seats of group $g$ in a row
> - $h$ - Safety distance between groups sitting next to each other

## Horizontal Social Distance

As the constraint is based on a rather complex set, you don't have to find it on your own.

$$\sum_{g \in \mathcal{G}} \sum_{\tilde{c} \in \tilde{\mathcal{C}}_{g,c}} X_{g,r,\tilde{c}} \leq 1 \quad \forall r \in \mathcal{R}, c \in \mathcal{C}$$

> **Note**
> At first glance, this constraint **looks rather easy**, but it is not - it is based on the set $\tilde{\mathcal{C}}_{g,c}$, a set not yet defined in the lecture.

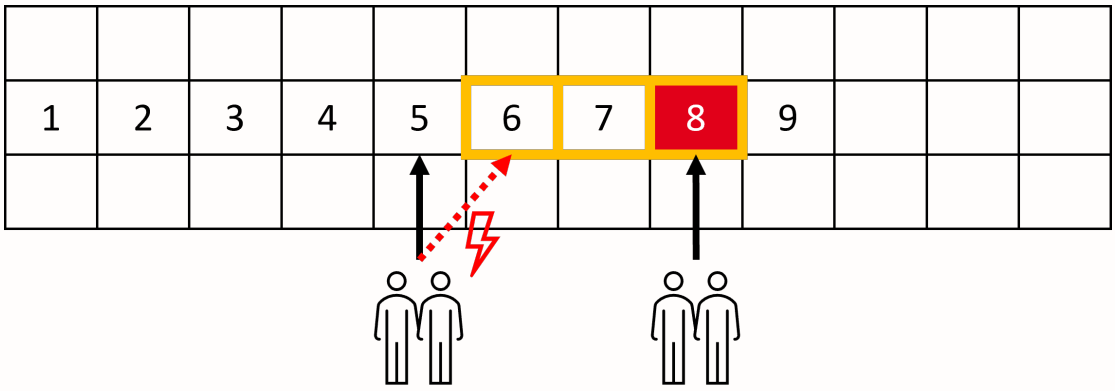
## The Social Distancing Set

$$\tilde{\mathcal{C}}_{g,c} = \left\{\tilde{c} \in \mathcal{C}_{g,r} | c - d_g + 1 - h \leq \tilde{c} \leq c\right\}$$

> **Remember:**
> - $d_g$ - Required seats of group $g$ in a row
> - $h$ - Safety distance between groups sitting next to each other
> - $\mathcal{C}_{g,r}$ - Feasible starting seats, as variables only exist for these seats (the row $r$ follows from the constraint)

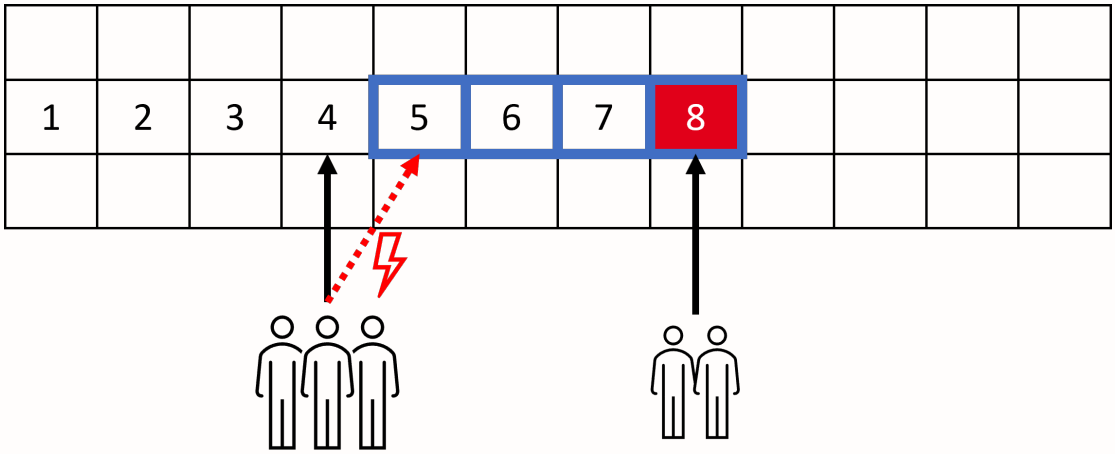
**Question: Can anybody explain the set?**

## Example: Two Groups



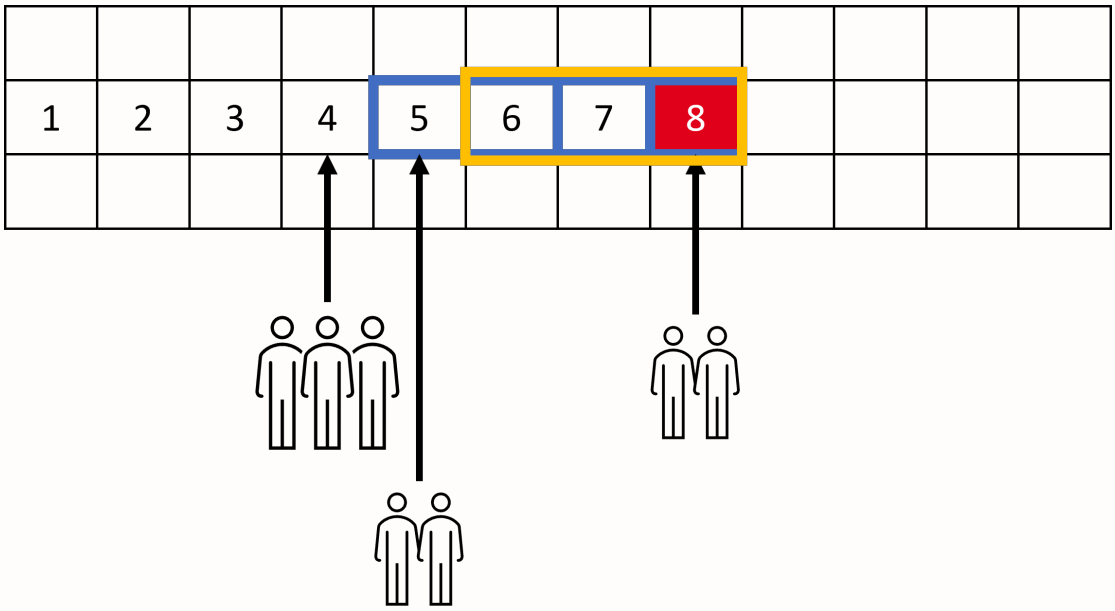

$$\underbrace{X_{1,2,\mathbf{6}} + X_{1,2,\mathbf{7}} + X_{1,2,\mathbf{8}}}_{g=1} + \underbrace{X_{2,2,\mathbf{6}} + X_{2,2,\mathbf{7}} + X_{2,2,\mathbf{8}}}_{g=2} \leq 1 \quad (r = 2, c = 8)$$

## Example: Different Group Sizes




$$\underbrace{X_{1,2,\mathbf{6}} + X_{1,2,\mathbf{7}} + X_{1,2,\mathbf{8}}}_{g=1} + \underbrace{X_{2,2,\mathbf{5}} + X_{2,2,\mathbf{6}} + X_{2,2,\mathbf{7}} + X_{2,2,\mathbf{8}}}_{g=2} \leq 1 \quad (r = 2, c = 8)$$

## Example: Three Groups




$$\underbrace{X_{1,2,\mathbf{6}} + X_{1,2,\mathbf{7}} + X_{1,2,\mathbf{8}}}_{g=1} + \underbrace{X_{2,2,\mathbf{6}} + X_{2,2,\mathbf{7}} + X_{2,2,\mathbf{8}}}_{g=2} + \underbrace{X_{3,2,\mathbf{5}} + X_{3,2,\mathbf{6}} + X_{3,2,\mathbf{7}} + X_{3,2,\mathbf{8}}}_{g=3} \leq 1 \quad (r = 2, c = 8)$$

Do you see

the pattern?

## Vertical Social Distance?

! **The goal of this constraint is to:**
Ensure that the vertical social distance is maintained between groups.

**We need the following:**
- $b$ - Vertical safety distance between groups
- $X_{g,r,c}$ - 1, if first left seat of $g$ is assigned to $r$ in $c$, else 0

## Vertical Social Distance

**Question: What could be the constraint?**

**Tip**
It is an extension of the horizontal social distance constraint we used before, but now we **block a rectangular area** instead of a single row.

$$\sum_{g \in \mathcal{G}} \sum_{\tilde{r} \in \tilde{\mathcal{R}}_r} \sum_{\tilde{c} \in \tilde{\mathcal{C}}_{g,c}} X_{g,\tilde{r},\tilde{c}} \leq 1 \quad \forall r \in \mathcal{R}, c \in \mathcal{C}$$

## Vertical Distance Set

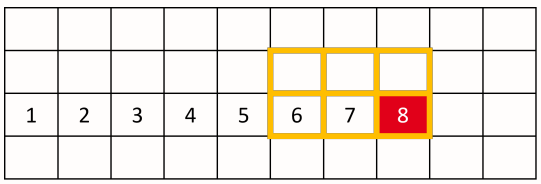
**Question: What could be the set?**

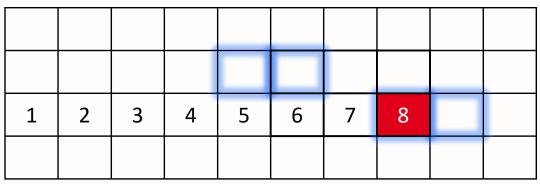
$$\tilde{\mathcal{R}}_r = \{\tilde{r} \in \mathcal{R} | r - b \leq \tilde{r} \leq r\}$$

**Note**
Remember:
- $b$ - Vertical safety distance between groups
- $X_{g,r,c}$ - 1, if first left seat of $g$ is assigned to $r$ in $c$, else 0

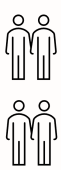
Let's look at an **example**.

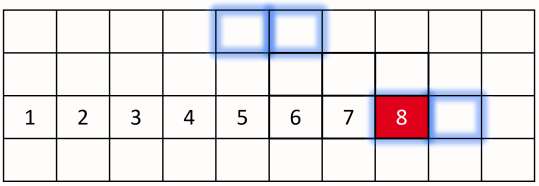
## Example: Two Groups




Example placement






- Yellow seats are **blocked by the group** in row 3 and column 8

- Blue allocations are possible (if second group has **size 2**)

## Arena Seating Problem

$$\text{Maximize} \quad \sum_{g \in \mathcal{G}} \sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} v_g \times X_{g,r,c}$$

subject to:

$$\sum_{r \in \mathcal{R}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq 1 \qquad \forall g \in \mathcal{G}$$

$$\sum_{g \in \mathcal{G}} \sum_{c \in \mathcal{C}_{g,r}} X_{g,r,c} \leq p_r \qquad \forall r \in \mathcal{R}$$

$$\sum_{g \in \mathcal{G}} \sum_{\tilde{r} \in \tilde{\mathcal{R}}_r} \sum_{\tilde{c} \in \tilde{\mathcal{C}}_{g,c}} X_{g,\tilde{r},\tilde{c}} \leq 1 \forall r \in \mathcal{R}, c \in \mathcal{C}$$

$$X_{g,r,c} \in \{0,1\} \qquad \forall g \in \mathcal{G}, r \in \mathcal{R}, c \in \mathcal{C}_{g,r}$$

## Why No Horizontal Constraint?

**Question: Why does the model contain only three constraints?**

- The vertical set $\tilde{\mathcal{R}}_r$ also contains $\tilde{r} = r$, the **row itself**
- For $\tilde{r} = r$, the rectangle constraint is **exactly** the horizontal one

> **Note**
> The rectangle constraint **subsumes** the horizontal social distance constraint. We can thus drop the horizontal constraint without losing anything.

# Model Characteristics

## Characteristics

**Questions: On model characteristics**

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?

> **Note**
> The model is **linear**, as the objective and all constraints are weighted sums of the variables. All variables are **binary**, making it a pure binary program.

## Model Assumptions

**Questions: On model assumptions**

- What assumptions have we made?
- Is our approach strategic or tactical/operational?

- Have we considered all social distancing constraints?
- What about aisle seats?
- Can you think of any other real-world constraints?

# In Practice

## Arena Seating Optimization

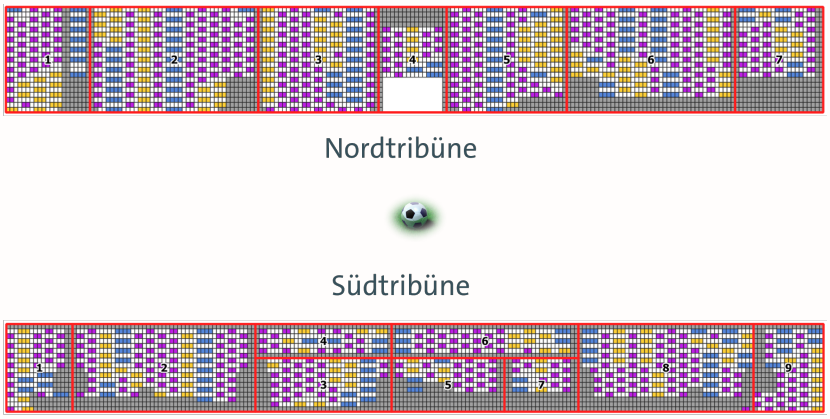
### Case study VfL Osnabrück

- Relegation Return Match in 2021
- 241 additional seats allocated (+12 percent)
- Compliance with all distancing requirements
- Approval from authorities

> ! **Important**
> Estimated additional revenue of **8,435 EUR** for one match.

### Seating Plan

| | Plätze | | |
|---|---|---|---|
| **Block** | **Verfügbar** | **Belegt** | **Belegt %** |
| Nord 1 | 349 | 135 | 38.7% |
| Nord 2 | 788 | 290 | 36.8% |
| Nord 3 | 586 | 217 | 37.0% |
| Nord 4 | 150 | 56 | 37.3% |
| Nord 5 | 549 | 204 | 37.2% |
| Nord 6 | 692 | 254 | 36.7% |
| Nord 7 | 266 | 99 | 37.2% |
| **Nord Gesamt** | **3380** | **1255** | **37.1%** |
| Süd 1 | 214 | 83 | 38.8% |
| Süd 2 | 587 | 218 | 37.1% |
| Süd 3 | 281 | 105 | 37.4% |
| Süd 4 | 165 | 61 | 37.0% |
| Süd 5 | 153 | 60 | 39.2% |
| Süd 6 | 225 | 83 | 36.9% |
| Süd 7 | 105 | 39 | 37.1% |
| Süd 8 | 576 | 213 | 37.0% |
| Süd 9 | 251 | 96 | 38.2% |
| **Süd Gesamt** | **2557** | **958** | **37.5%** |
| **Nord+Süd Gesamt** | **5937** | **2213** | **37.3%** |




Modellrechnung Bremer Brücke

### Related Work

- The model of this lecture is published in U. Dkaidik and M. Koch [1]

**Similar studies** have been conducted globally:

- Music Hall Eindhoven: D. Blom, R. Pendavingh, and F. Spieksma [2]
- US College-level venues, e.g. Football, Basketball, Hockey
- Safe Seating Solutions platform
- General 2D-Knapsack applications

# Conclusion

## Optimization Benefits

- Optimization enables **rapid generation and evaluation**
- We can easily adapt to **various distancing requirements**:
  - Horizontal and vertical spacing between groups
  - Groups per row limits
  - Aisle seat restrictions
  - Group size constraints
  - Multi-row group allocation

## Wrap Up

- **Revenue optimization** through applied optimization
- **Increased participant capacity** vs basic approaches
- Flexible adaptation to **various distancing requirements**
- Can be adapted easily to **any seating requirements**

> **And that's it for today's lecture!**
> We now have covered the arena seating problem based on a real-world application and are ready to start solving the corresponding tasks in the upcoming tutorial.

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Questions?

# Literature

- Press coverage of the VfL Osnabrück case: University of Hamburg newsroom (in German)

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

[1] U. Dkaidik and M. Koch, "Arena Seat Planning Under Distancing Rules," in *Operations Research Proceedings 2024*, Cham: Springer, 2025, pp. 82–88. doi: 10.1007/978-3-031-92575-7_12.

[2] D. Blom, R. Pendavingh, and F. Spieksma, "Filling a Theater During the COVID-19 Pandemic," *INFORMS Journal on Applied Analytics*, vol. 52, no. 6, pp. 473–484, 2022, doi: 10.1287/inte.2021.1104.